



HPE 3par Training Module



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Agenda

- ▶ **HPE 3PAR Solution Overview**
- ▶ **Hardware Overview**
- ▶ **Management Console GUI and CLI Introduction**
- ▶ **Storage Concepts and Terminology**
- ▶ **Storage Configuration**
- ▶ **Host Connectivity and Storage Allocation**
- ▶ **Local Replication: Virtual Copy and Clones**
- ▶ **Dynamic Optimization**
- ▶ **Priority Optimization**



Agenda

- ▶ **Virtual Domains**
- ▶ **Adaptive Optimization**
- ▶ **Remote Copy (RC)**
- ▶ **3Par capacity Management & Troubleshooting**





HPE 3par Solutions Overview



The world has changed

and storage must change with it



Explosive growth
& new workloads



Virtualization
& automation



Cloud & utility
computing



Infrastructure &
technology shifts

Customers tell us storage is:

Too complicated to manage

Expensive & hard to scale

Isolated & disconnected

Inefficient & inflexible

Storage needs to be:

Simple

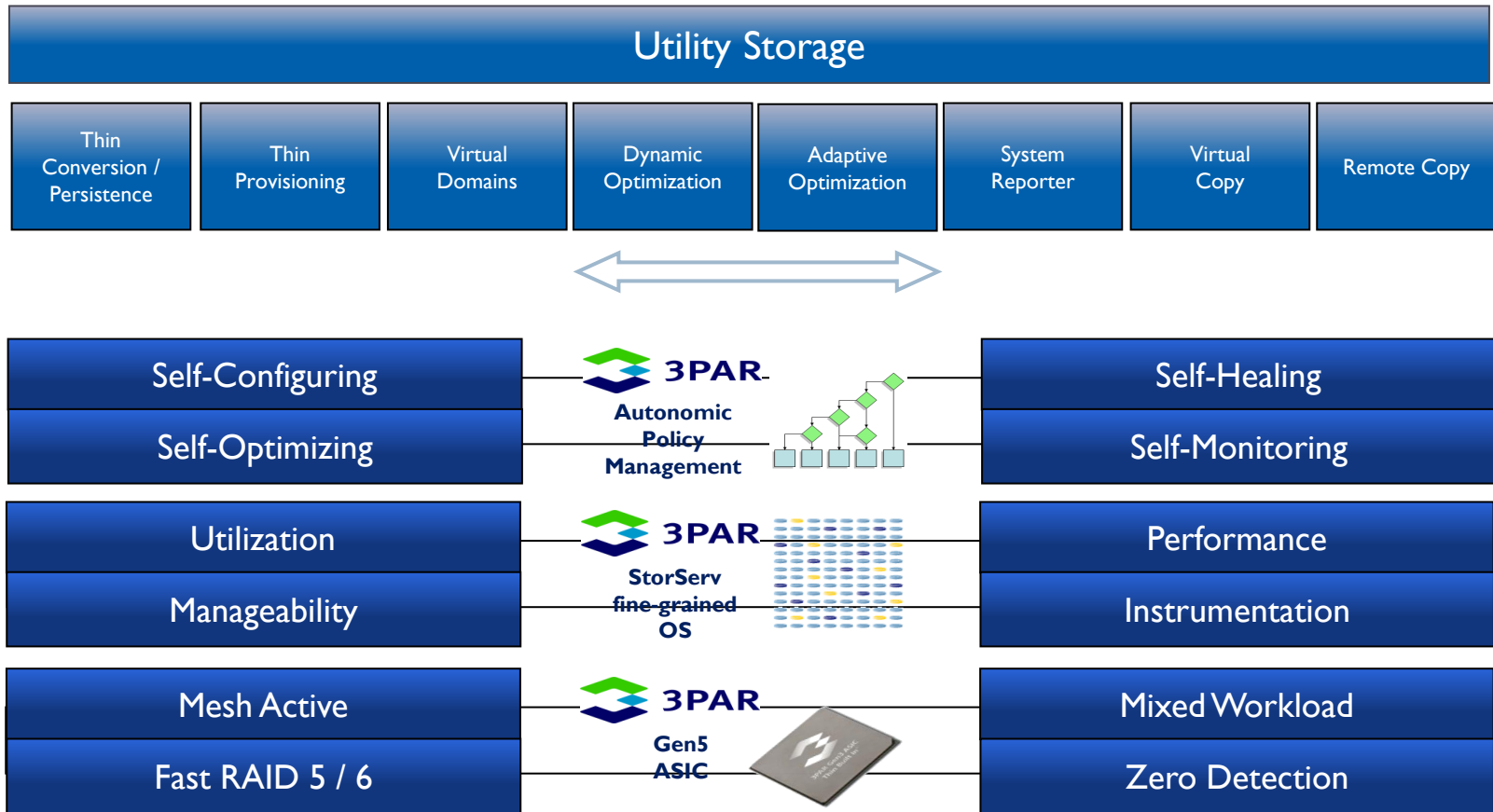
Scalable

Smart

Self-Optimized



Architectural differentiation: Purpose built



HP Enterprise Storage

Simplified Portfolio and Branding

3PAR StoreServ



Utility Storage
SMB to Enterprise

StoreVirtual VSA



Scale-Out
SAN SMB

StoreEasy



Filesystem Storage
SMB to Enterprise

StoreOnce



De-Duplication
SMB to Enterprise



HPE Storage array positioning

The industry's broadest and deepest flash portfolio



MSA

Affordable acceleration



SimpliVity

Enterprise hyper converged infrastructure (HCI)



Nimble Storage

Simple and predictive flash for new-style deployments



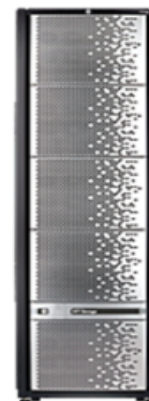
3PAR StoreServ

Flexible Tier-1 features for the all-flash data center



Primera

Flexible Tier-0 features for the all-flash data center



XP7

For extreme data availability needs

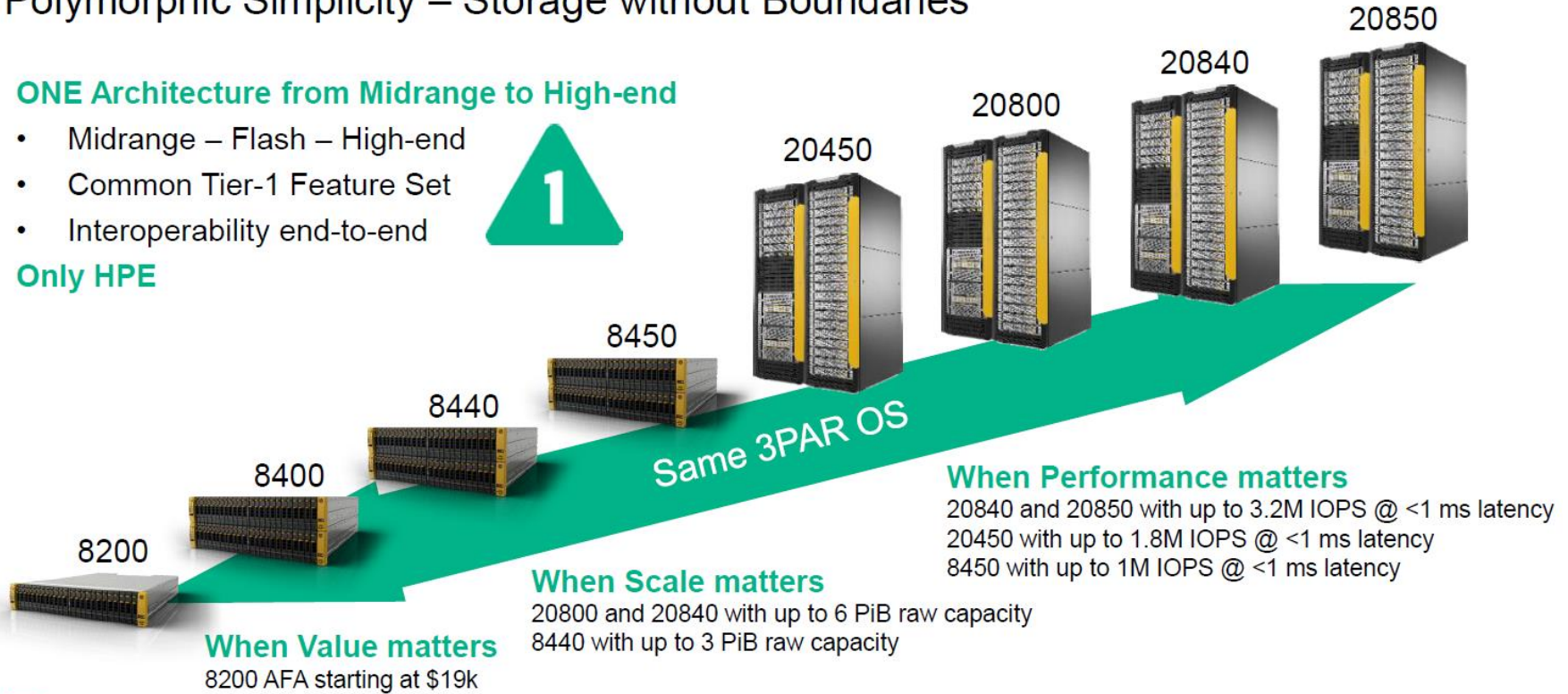
Eliminating distinctions between Midrange and Tier 1

Polymorphic Simplicity – Storage without Boundaries

ONE Architecture from Midrange to High-end

- Midrange – Flash – High-end
- Common Tier-1 Feature Set
- Interoperability end-to-end

Only HPE



3PAR Industry Leadership

Best storage technology in the market – built for virtual environments

3PAR Federation Technologies

3PAR Flash Optimization

3PAR Optimization and Compaction Technologies

3PAR Converged Block, File and Object Access

3PAR Thin Technologies

3PAR Persistent Technologies

3PAR Mesh-Active Architecture



The 3PAR StoreServ Family



	8200	8400	8450	8440	20450	20800	20840	20850
Controller Nodes	2	2 - 4	2 - 4	2 - 4	2 - 4	2 - 8	2 - 8	2 - 8
16 Gb Fibre Channel Ports	4 - 12	4 - 24	4 - 24	4 - 24	0 - 80	0 - 160	0 - 160	0 - 160
10Gb iSCSI / FCoE Ports	0 - 4	0 - 8	0 - 8	0 - 8	0 - 40	0 - 80	0 - 80	0 - 80
1Gb Ethernet IP Ports	0 - 8	0 - 16	0 - 16	0 - 16	NA	NA	NA	NA
10Gb Ethernet IP Ports	0 - 4	0 - 8	0 - 8	0 - 8	0 - 24	0 - 48	0 - 48	0 - 48
Built-in IP Remote Copy Ports	2	2 - 4	2 - 4	2 - 4	2 - 4	2 - 8	2 - 8	2 - 8
Cache per node-pair / per system GiB	64 / 64	64 / 128	192 / 384	192 / 384	896 / 1792	448 / 1792	900 / 3600	896 / 3584
Flash Cache per Node-Pair / system GiB	768 / 768	768 / 1536	NA	4000 / 8000	NA	8192 / 32768	12288 / 49152	NA
Drives per StoreServ array	8 - 240	8 - 576	8 - 480	8 - 960	8 - 512	8 - 1920	8 - 1920	8 - 1024
Available Drive Types > Go to the 8000 drive overview > Go to the 20000 drive overview	SSD							
	15k SAS	yes	yes	yes	yes	yes	yes	yes
	10k SAS	yes	yes	NA	yes	NA	yes	NA
	7.2k NL SAS	yes	yes	NA	yes	NA	yes	NA
	SAS	yes	yes	NA	yes	NA	yes	NA
Max SSD per StoreServ	120	240	480	480	512	1024	1024	1024
Max Raw Capacity (TiB)	750	2400	1843	3000	1966	6000	6000	3932
Max random IOPS front-end performance ¹	Distributed	342'000	659'000	786'000	786'000	933'000	1'379'000	1'724'000
	Local Node	400'000	800'000	1'000'000	1'000'000	1'800'000	2'500'000	3'200'000
Max sequential GB/s front-end performance ²	Distributed	6.1	12.6	12.6	12.6	12.3	30.3	30.3
	Local Node	11.6	23.2	24	24	38	75	75

3PAR Components

Drive Chassis (4U)

- Capacity building block
 - Drive Magazines
- Add non-disruptively
- Industry leading density

Controller Nodes (4U)

- Performance and connectivity building block
 - Adapter cards
- Add non-disruptively
- Runs independent OS instance

Full-mesh Back-plane

- Post-switch architecture
- High performance, tightly coupled
- Completely passive

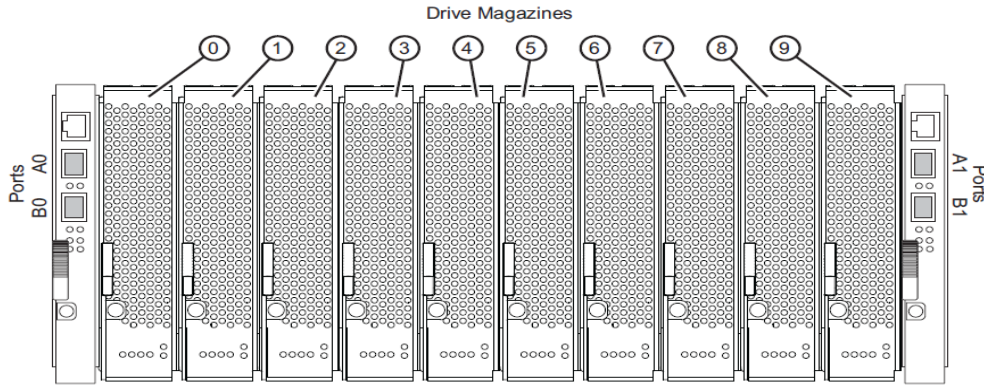


3PAR 40U, 19" Cabinet
Built-In Cable
Management

Service Processor (IU)

- Remote error detection
- Supports diagnostics and maintenance
- Reporting to 3PAR Central

3PAR Architecture: Drive Chassis (DC4)

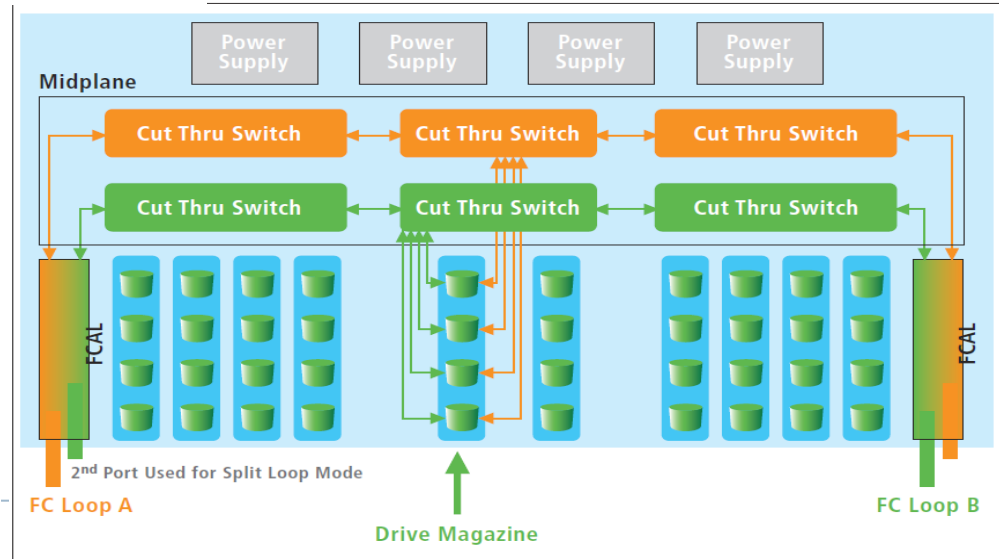


DC4 Drive Chassis or “Cage” Contains

- 10 Drive bays
- Each bay holds one magazine
- Each magazine holds 4 drives.

Disk Cage is

- (1+1) redundant power supplies
- Dual FCAL loops
- Dual switches for dual ported drives



3PAR Architecture: Drive Chassis (DC3)

I 'magazine'

- all drives must be of same type



Drive Chassis or “cage” contains 4 drive bays that accommodate:

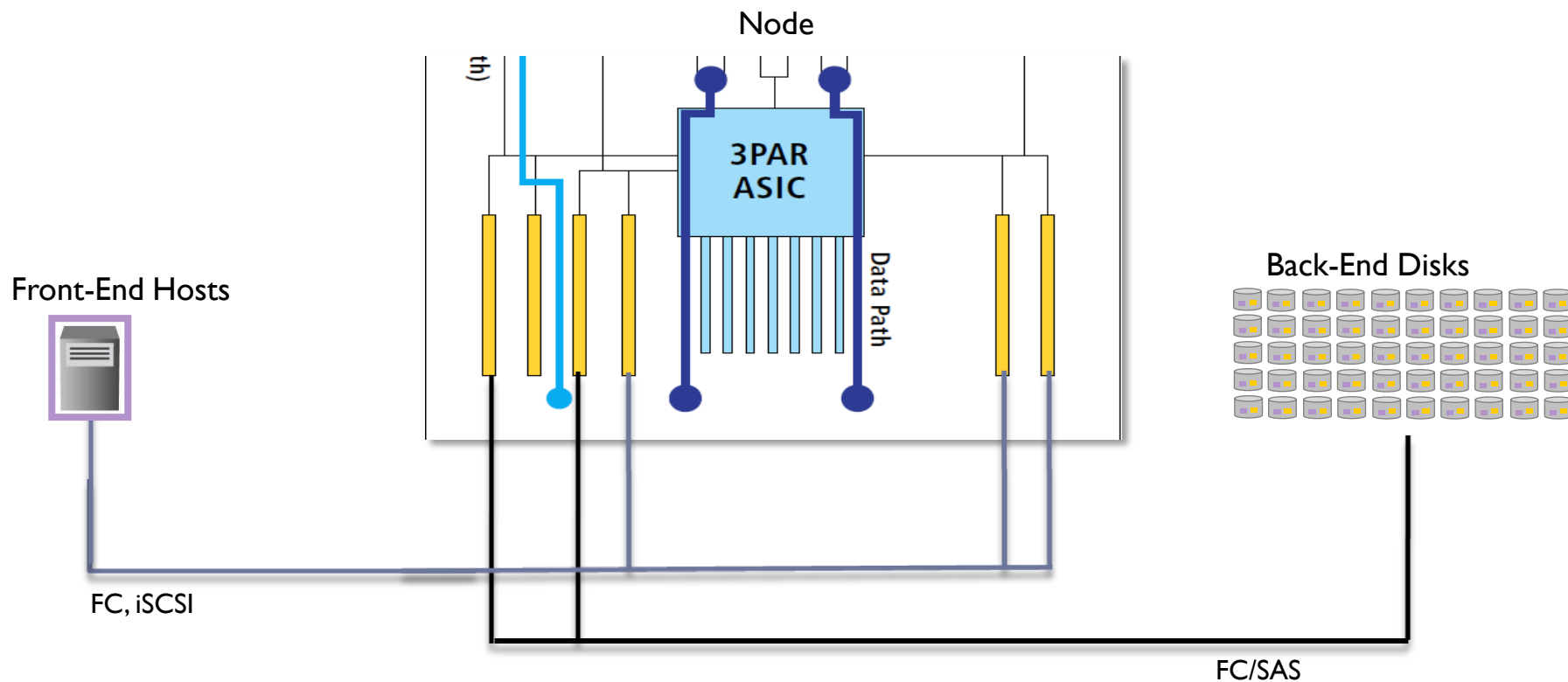
- ▶ 4 drive magazines
- ▶ Each magazine holds four disks
- ▶ Each disk is individually accessible



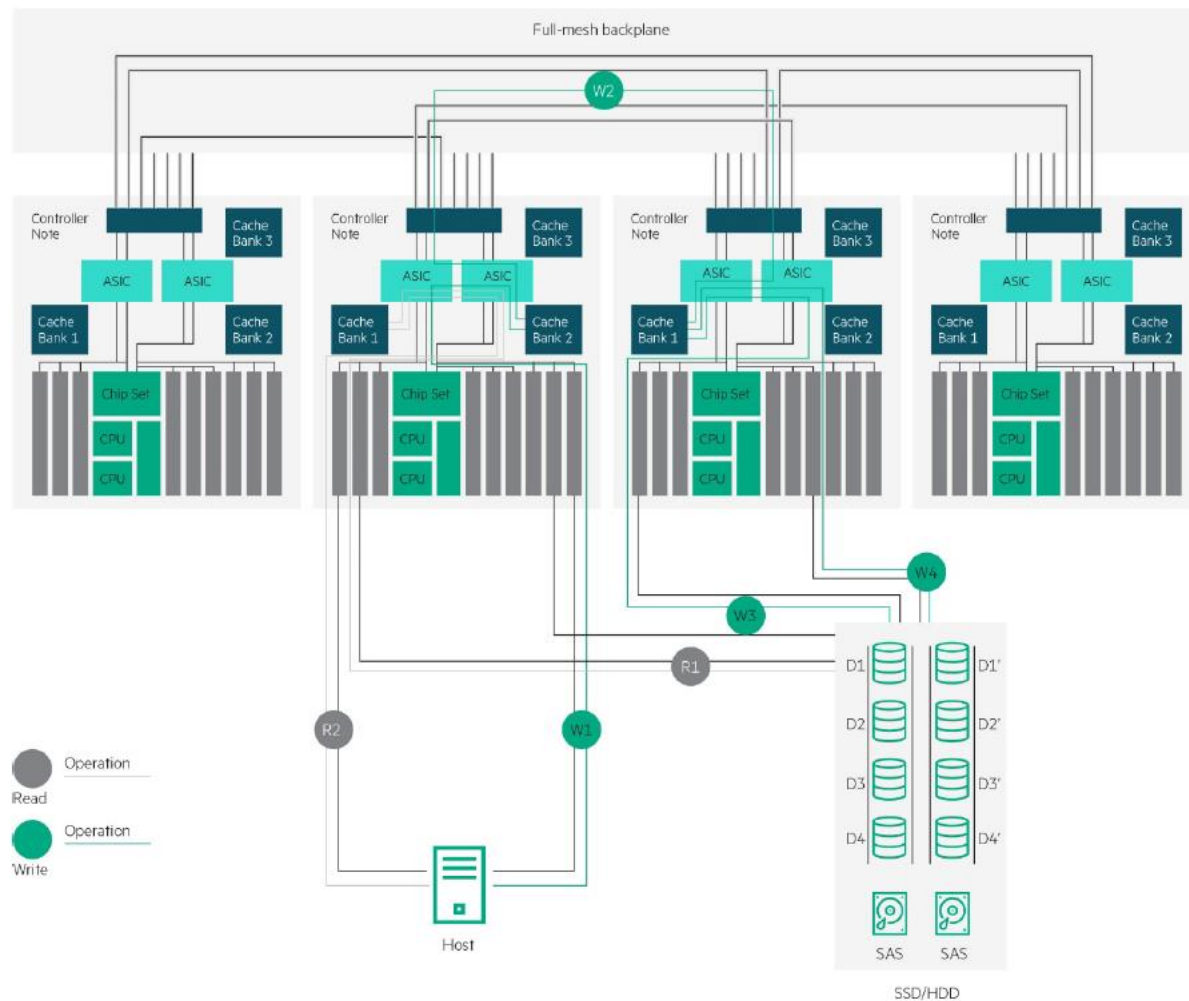
Connectivity: Host to Node/ASIC to Drive

3PAR StorServ Connectivity

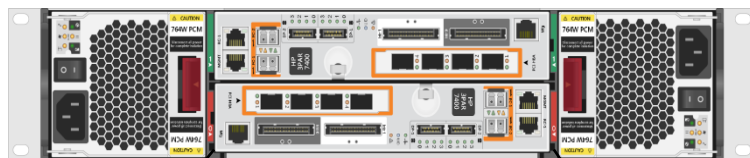
Multi – Protocol Connectivity



Host W/R



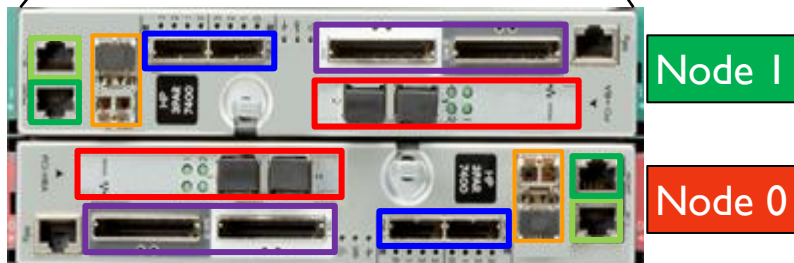
3PAR StoreServ controller enclosure



Rear view



Front view



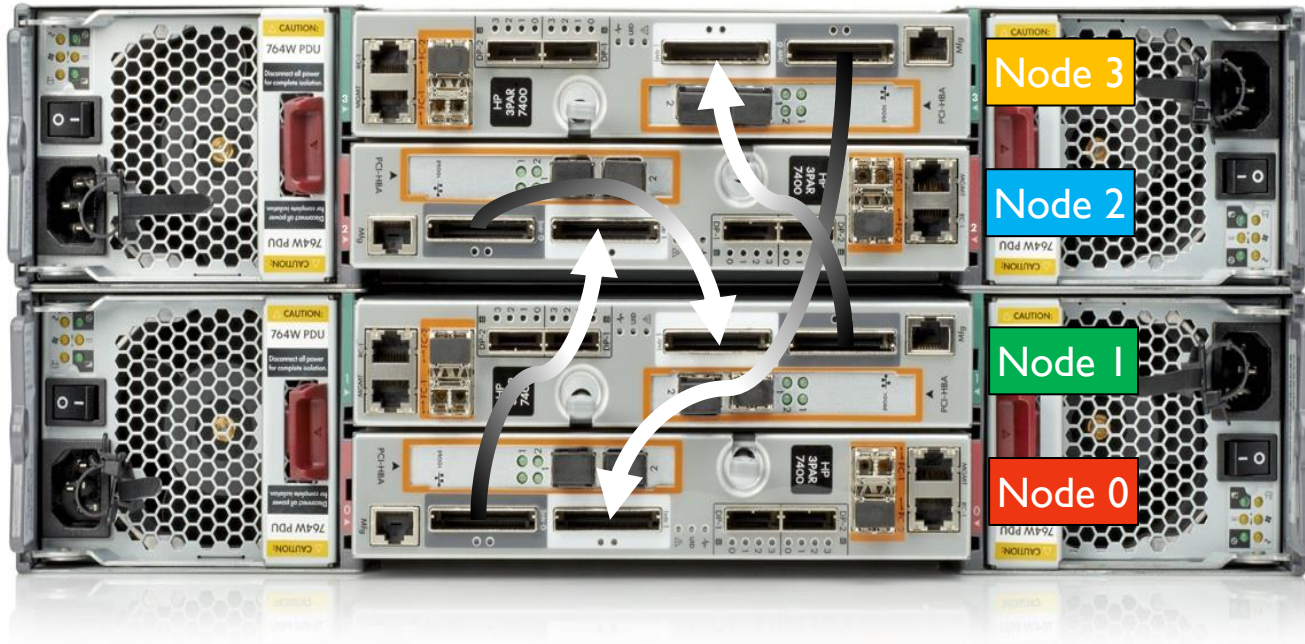
Node 1

Node 0

-  Built-in Eth Remote Copy Port
-  Eth Mgmt. Port
-  2x built-in 8Gbit/s FC
-  2x 4-lane 6Gbit/s SAS for drive chassis connection
-  Optional PCIe card
 - 4x 8GB/s HBA or
 - 2x 10Gbit/s CNA
-  Controller interconnect

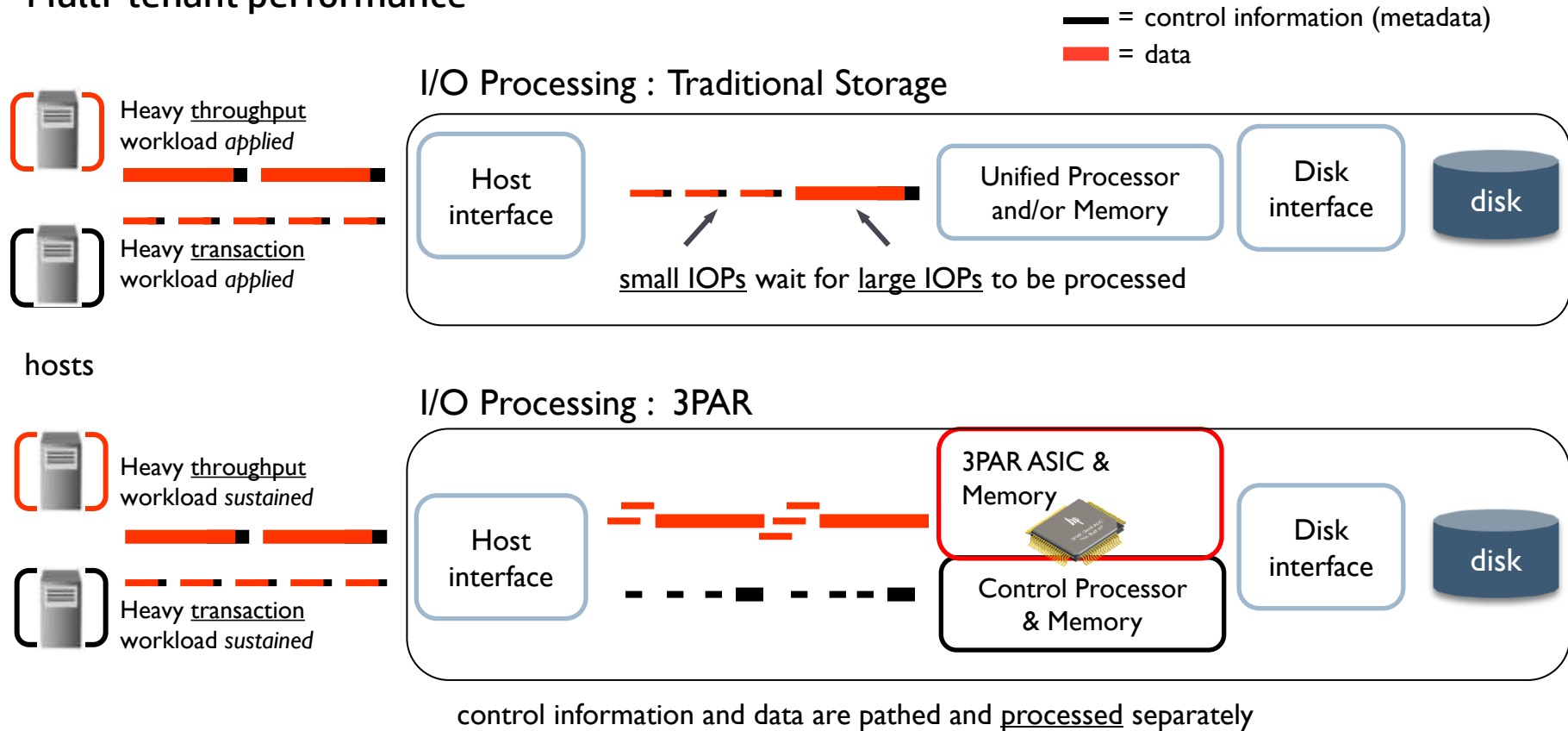
3PAR StoreServ 4-node System

Controller Interconnect



3PAR Mixed workload support

Multi-tenant performance



3PAR disk chassis

Mix and match drives and shelves as required

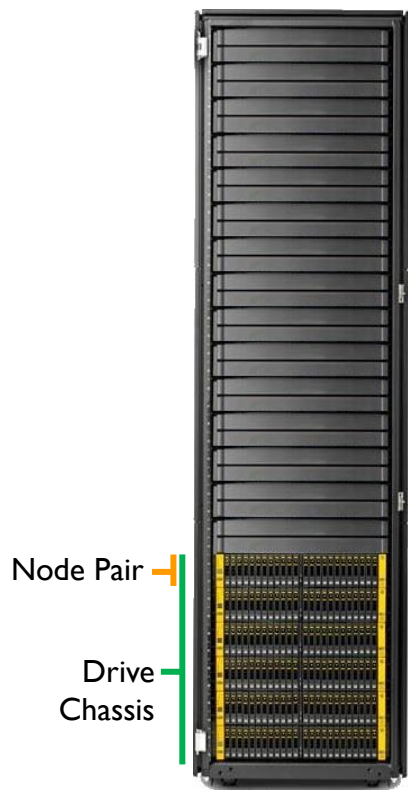
2U with 24 SFF drive slots



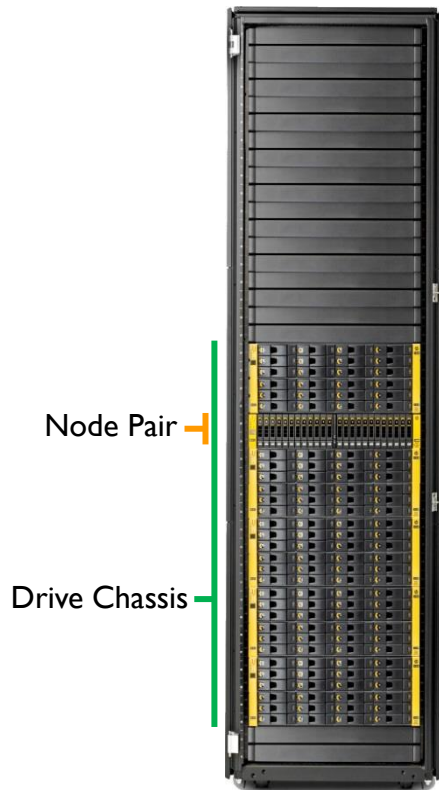
4U with 24 LFF drive slots



3PAR max. Configurations



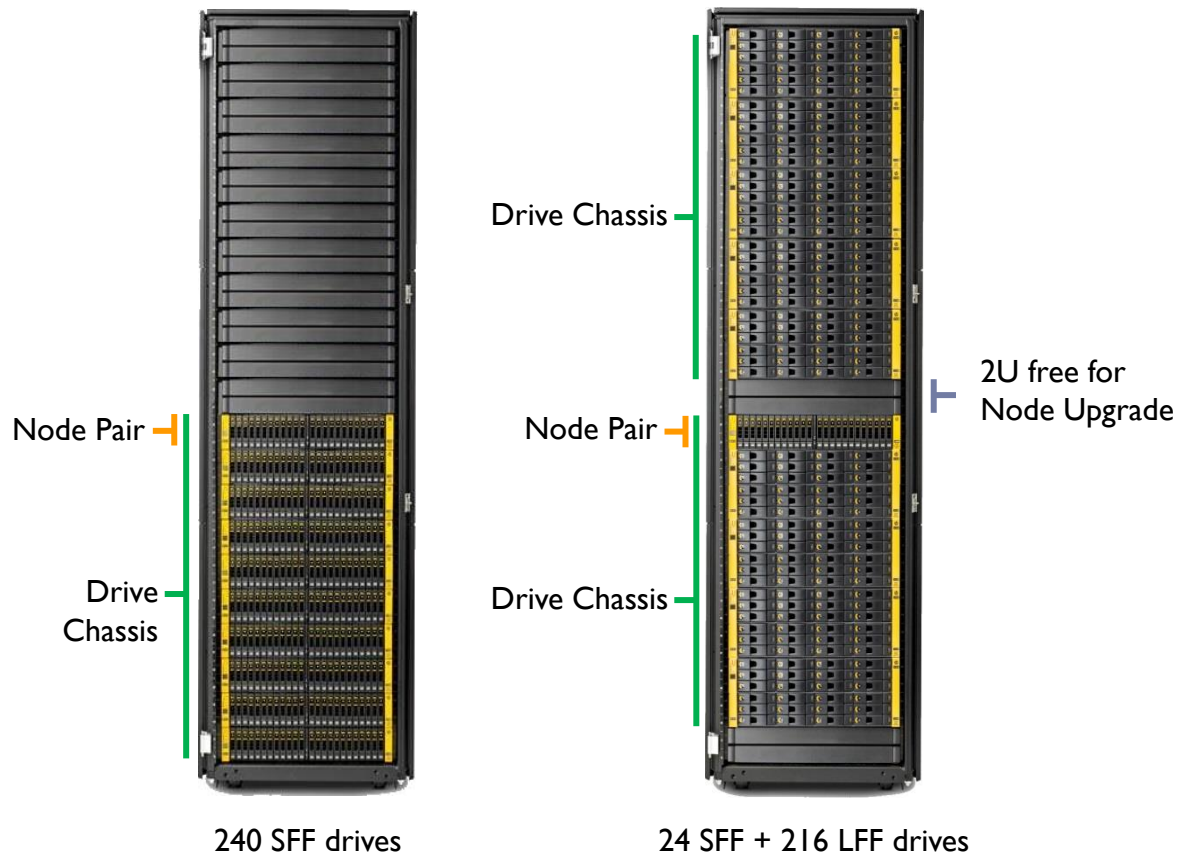
144 SFF drives



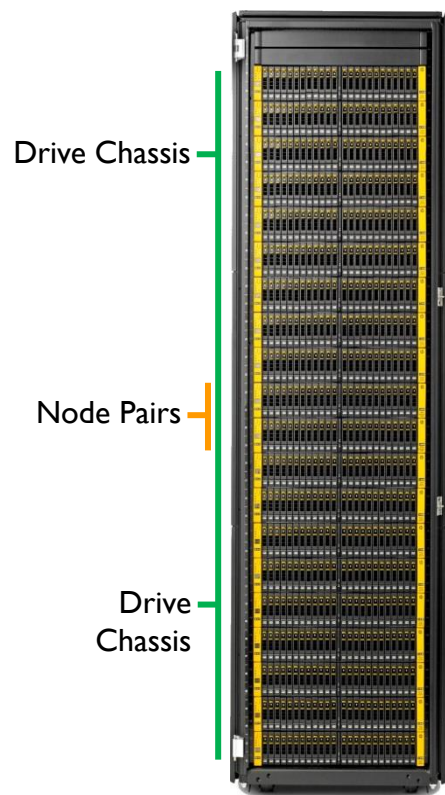
24 SFF + 120 LFF drives



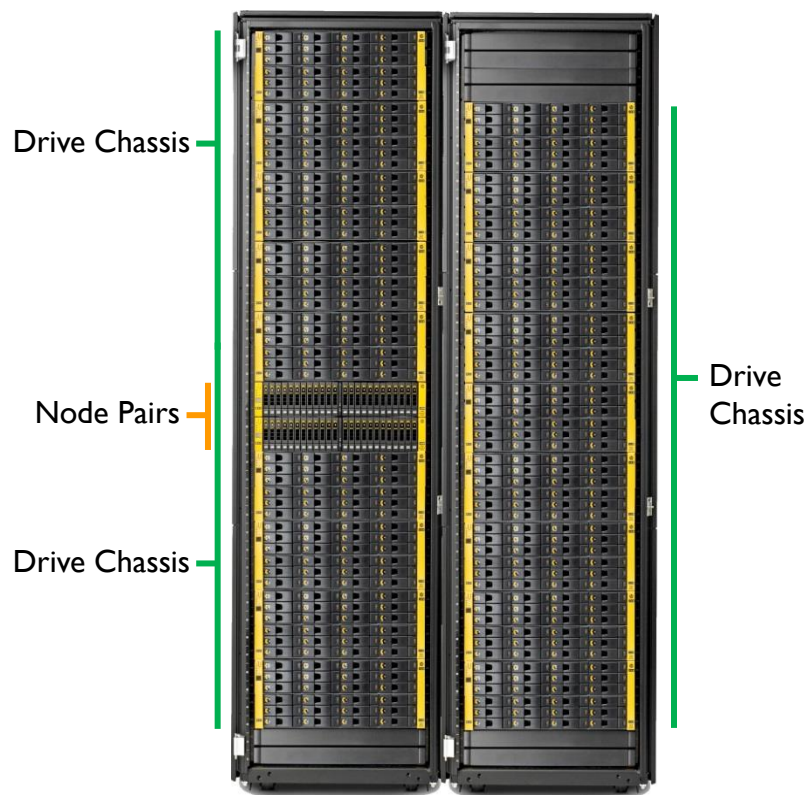
3PAR 2-node max Configurations



3PAR 4-node max Configurations



480 SFF drives

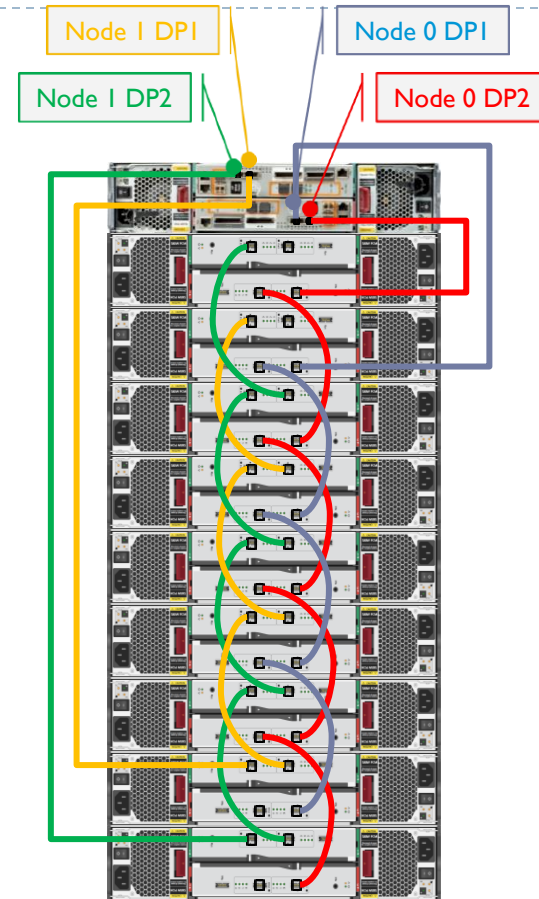


48 SFF + 432 LFF drives



HP 3PAR 7000 SAS Backend cabling

- Each node has 2 SAS controller-to-enclosure connectors with 4 x 6Gbit/s lanes each
- Controller enclosure internal drives are connected to controller port DP1
- First external enclosure is added to controller ports DP-2
- Additional enclosures then alternate between controller ports DP-1 and DP-2
- Each drive enclosure comes with mounting rail kit, 2 x IO-modules, 2 x power cables and 2 x 1m SAS cables.
- Enclosures may be installed in adjacent racks. 2m and 6m SAS expansion cable kits are available



HP 3PAR StoreServ Hardware Building Blocks

Base Storage Systems



HP 3PAR StoreServ 7200
(2 nodes, 4 FC ports, 24 SFF slots)



HP 3PAR StoreServ 7400
(2-node, 4 FC ports, 24 SFF slots)



HP 3PAR StoreServ 7400
(4-node, 8 FC ports, 48 SFF slots)

Host Adapters



4-port 8Gb/s
FC HBA



2-port 10Gb/s
iSCSI/FCoE CNA

Expansion Drive Enclosures



HP M6710 2.5in 2U SAS



HP M6720 3.5in 4U SAS

Drives



SFF SAS
HDD & SSD



LFF SAS
HDD &
SSDs

Racks



HP G3 rack



Customer-supplied rack
(4-post, square hole,
EIA standard, 19 in.,
rack from HP or
other suppliers)

Service Processor



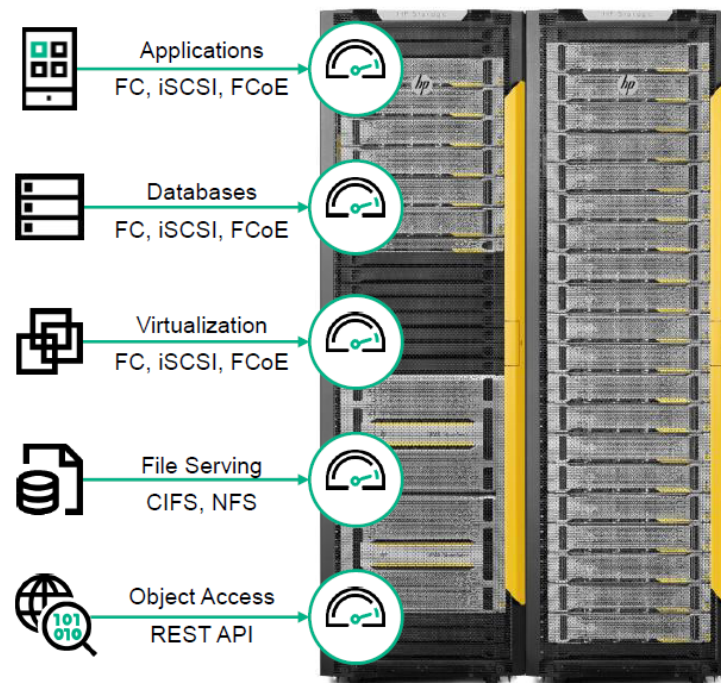
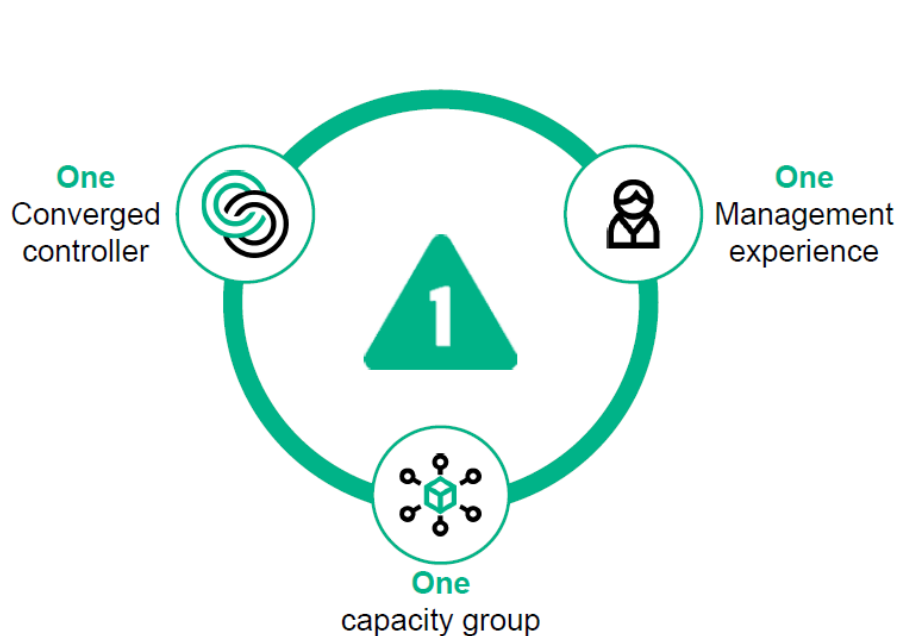
VM-based
(Default)



Physical
(Optional)

Converged Block, File, and Object Access

For the 7000, 8000 and 20000 family





Software & Features



3PAR 8000 and 20000 Software Title Details



Replication SW Suite

- Virtual Copy (VC)
- Remote Copy (RC)
- Peer Persistence (PP)
- Cluster Extension Windows (CLX)



Data Optimization SW Suite

- Dynamic Optimization
- Adaptive Optimization
- Peer Motion
- Priority Optimization



Security SW Suite

- Virtual Domains
- Virtual Lock



- System Reporter
- 3PARInfo
- Online Import license (1 year)
- System Tuner
- Host Explorer
- Multi Path IO SW
- VSS Provider
- Scheduler



File Persona SW Suite



Data Encryption



Policy Server

- Policy Manager Software



Smart SAN for 3PAR

vmware

Recover Manager Central for vSphere



Application SW Suite for Oracle



Application SW Suite for MS SQL



Application SW Suite for MS Exchange



Application SW Suite for MS Hyper-V

3PAR Operating System SW Suite

- Rapid Provisioning
- Autonomic Groups
- Autonomic Replication Groups
- Autonomic Rebalance
- LDAP Support
- Access Guard
- Host Personas

- Adaptive Flash Cache
- Persistent Cache
- Persistent Ports
- Management Console
- Web Services API
- SMI-S
- Real Time Performance Monitor

- Full Copy
- Thin Provisioning
- Thin Copy Reclamation
- Thin Persistence
- Thin Conversion
- Thin Deduplication for SSD
- 3PAR OS Administration Tools
 - CLI client
 - SNMP

Management – 3PAR CLI and REST API

Dashboard: **Health, Performance, Capacity at a**

Aggregation of information from up to 32 HPE 3PAR StoreServ arrays

Search: **Instantly find what you want**

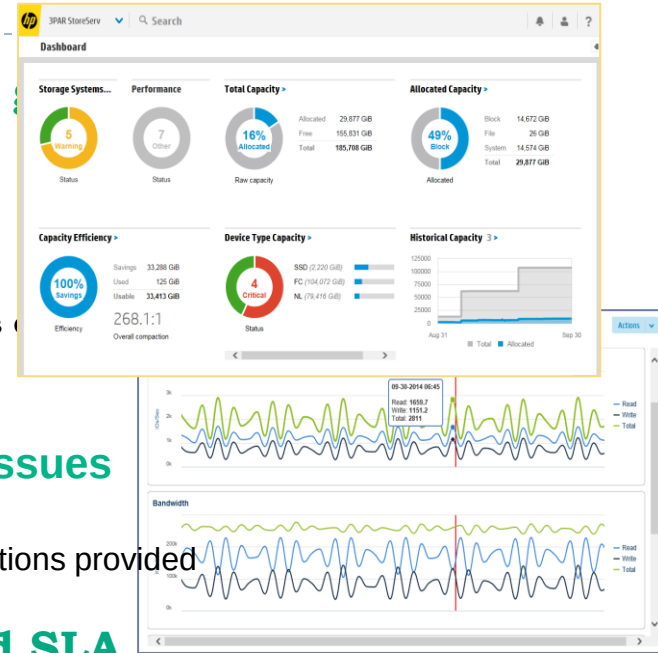
Find objects in seconds with intelligent search that remembers previous

Proactive tools: **Identify capacity and performance issues early**

1-click templates to assess system performance health with remediation actions provided

Federation: **Run workloads at the right cost and SLA**

One-click data mobility via Storage Federation



The entire HPE 3PAR StoreServ portfolio managed through one single console (SSMC)



HPE 3par HA Architecture

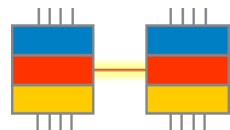


Legacy Storage Hardware Architecture

Traditional Tradeoffs

Traditional Modular Storage

Host Connectivity

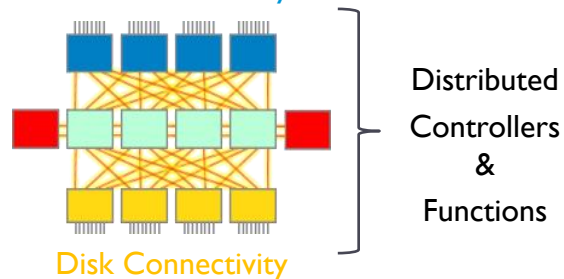


Disk Connectivity

Cost-efficient usually active-passive dual-controller design limited in scalability and resiliency.

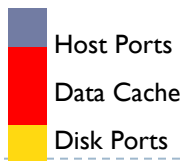
Traditional Monolithic Storage

Host Connectivity



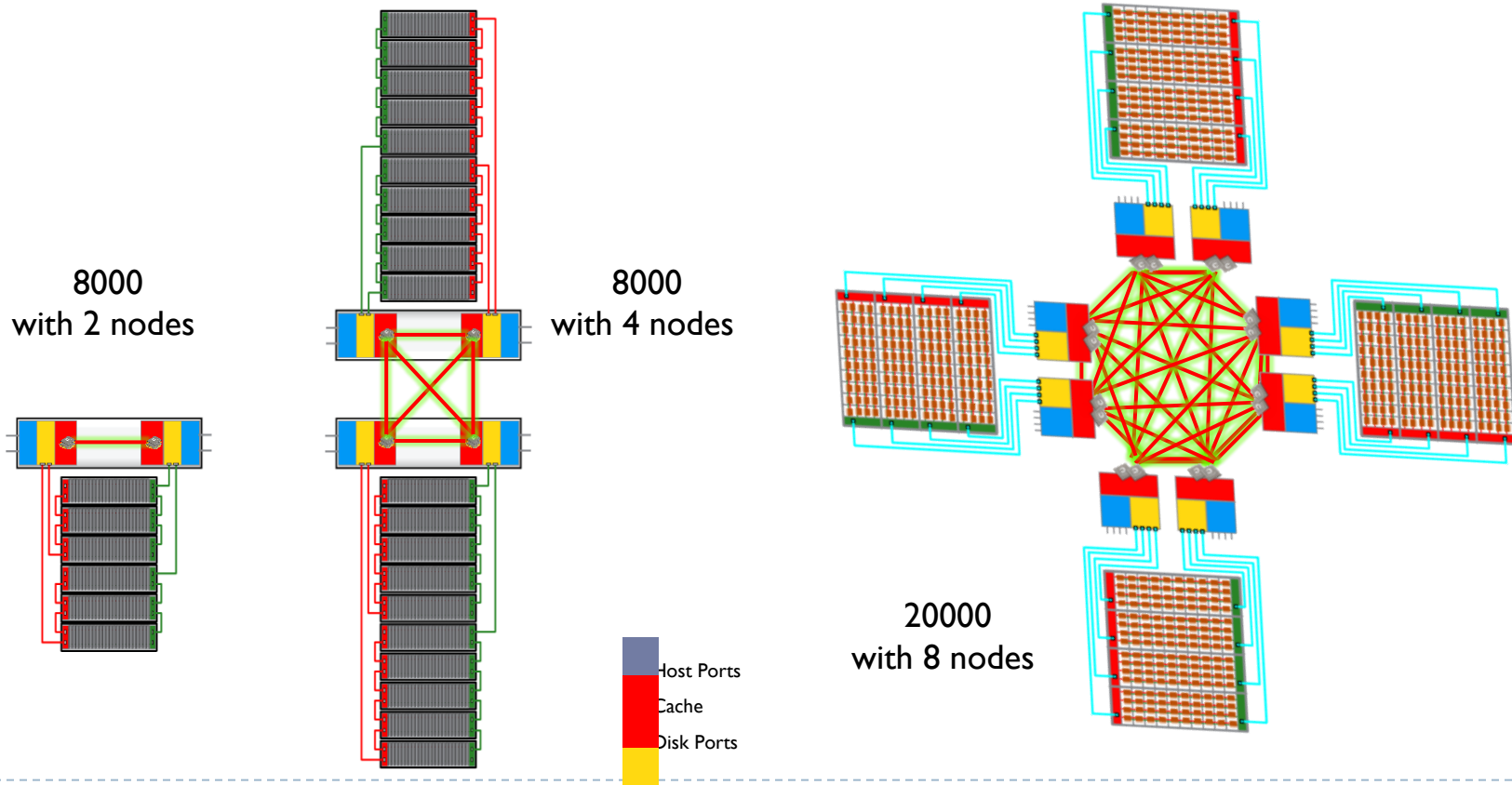
Disk Connectivity

Scalable, resilient and active-active but costly.
Might not meet multi-tenant requirements efficiently



3PAR Hardware Architecture

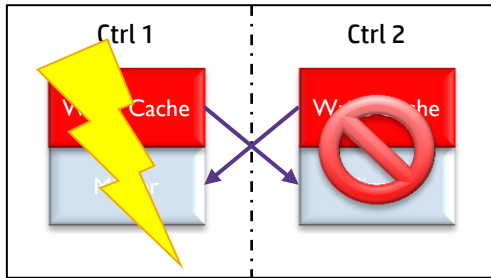
Cost-effective, scalable, resilient, meshed, active-active



HP 3PAR High Availability

Write Cache Re-Mirroring

Traditional Mid-range Arrays

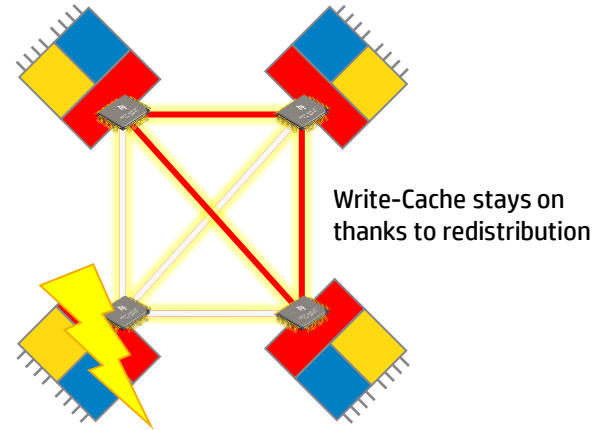


Write-Cache off for data security

Traditional Write-Cache Mirroring

Either poor performance due to write-thru mode
or risk of write data loss

3PAR StoreServ

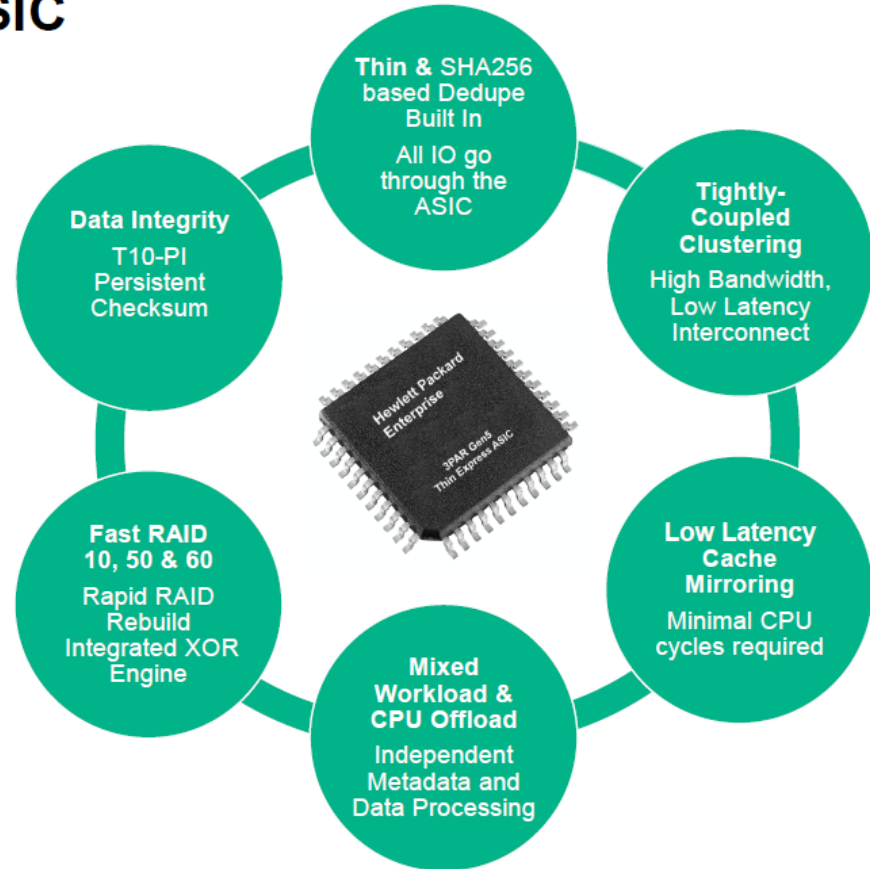
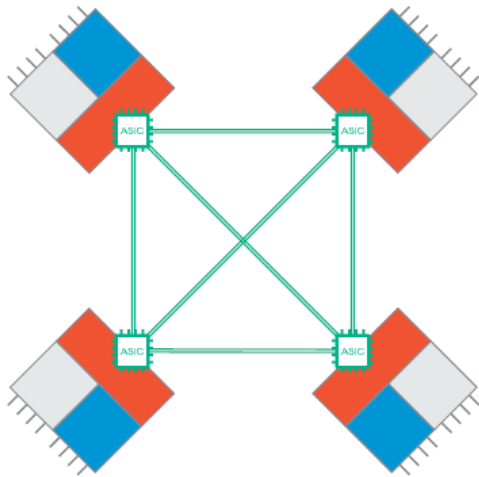


Persistent Write-Cache Mirroring

- No write-thru mode – consistent performance
- Works with 4 and more nodes

3PAR Gen5 Thin Express ASIC

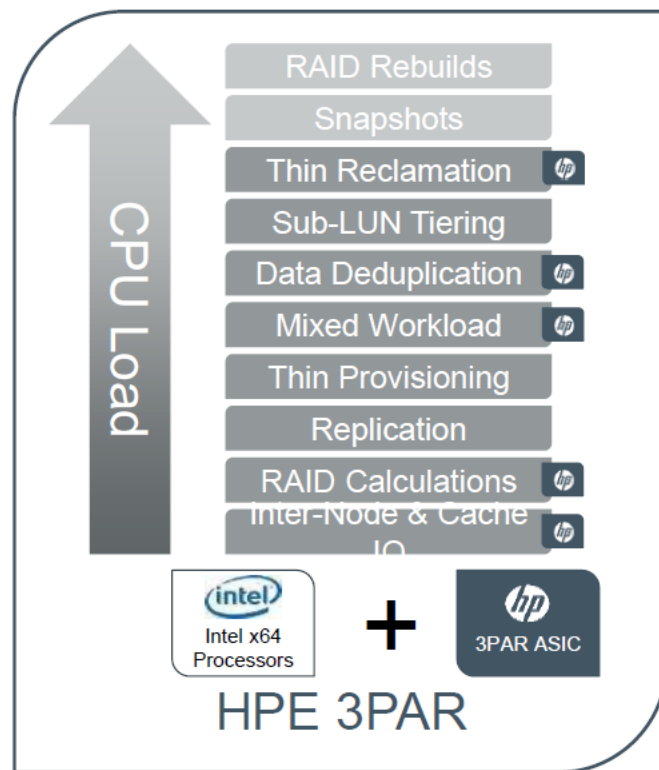
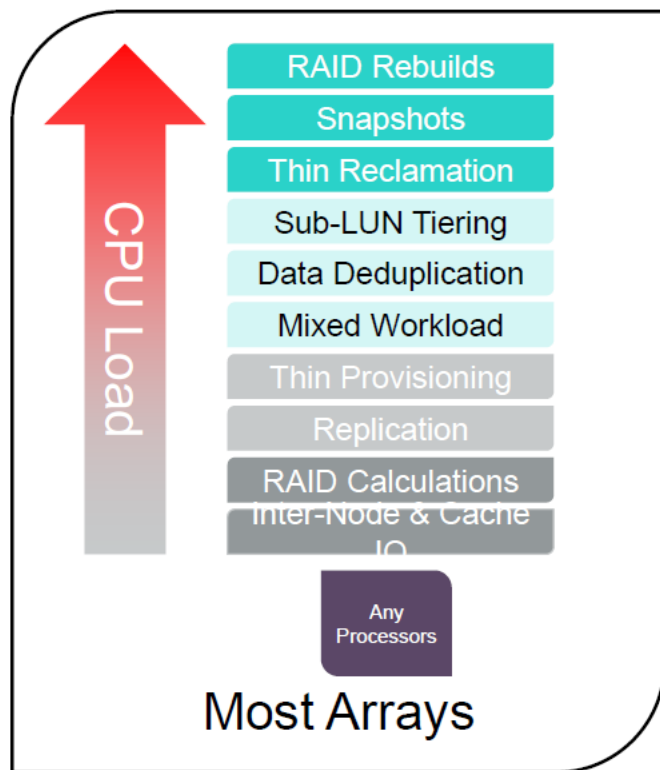
The Heart of every 3PAR



Watch the "Understand the 3PAR ASIC" video: <http://www.youtube.com/watch?v=LmQMqWNGAZE>

HPE 3PAR ASIC

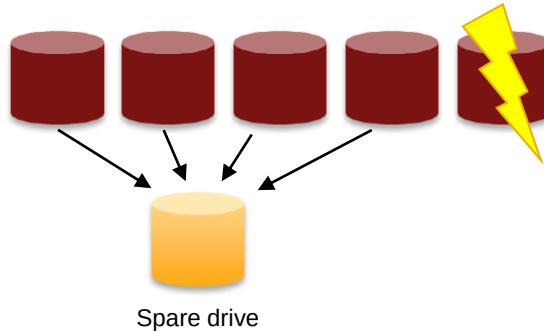
It makes the difference in real-world environments



HP 3PAR High Availability

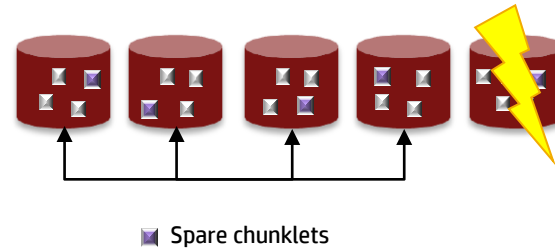
Spare Disk Drives vs. Distributed Sparing

Traditional Arrays



Few-to-one rebuild
hotspots & long rebuild exposure

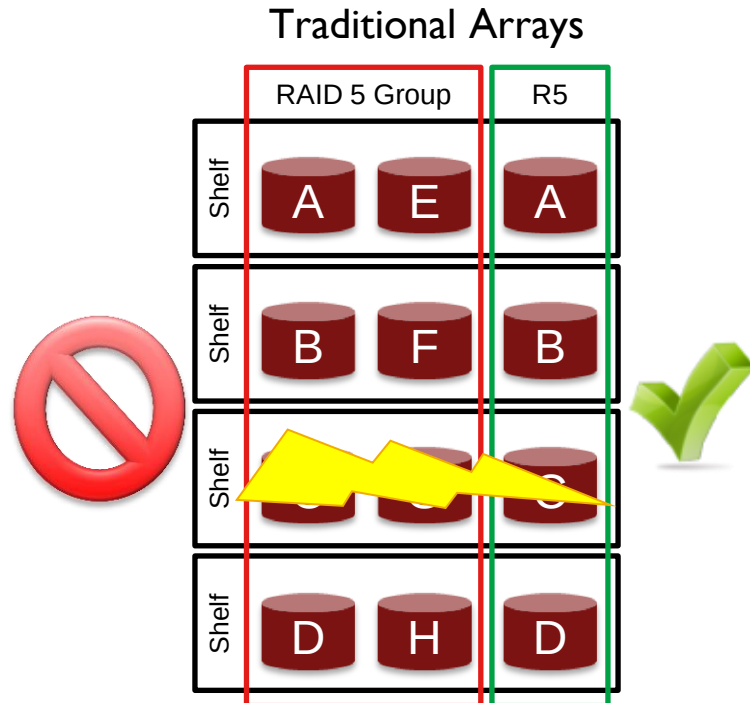
3PAR StoreServ



Many-to-many rebuild
parallel rebuilds in less time

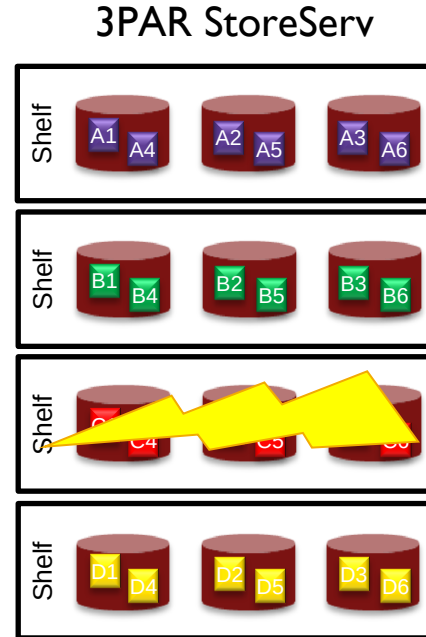
HP 3PAR High Availability

Guaranteed Drive Shelf Availability if desired



Shelf-dependent RAID

Shelf failure might mean no access to data

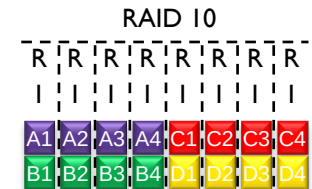
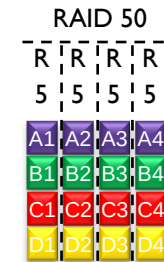


Shelf-independent RAID

Raidlet Groups for any RAID level

Data access preserved with HA Enclosure

User selectable per CPG



HA Enclosure vs. HA Drive

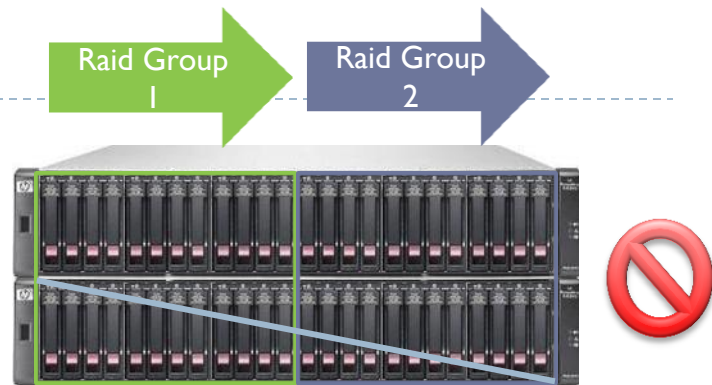
Tier 1 Availability feature in a modular class array

- ▶ Most modular arrays stripe Data and Parity across Enclosures.

If you lose access to ≥ 2 drives in an Enclosure you lose access to **ALL Data** in the array

Enterprise Arrays, including the 3PAR StoreServ 7000 allow configurations that stripe Data and Parity across Enclosures

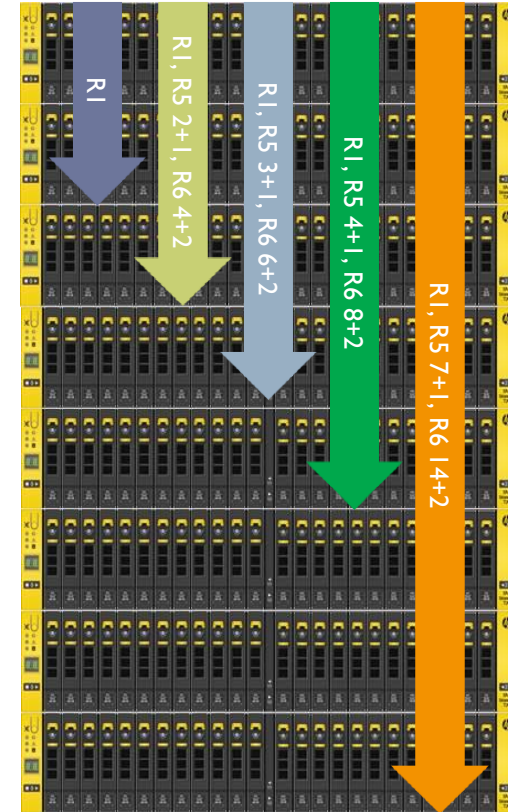
In the event of a **Entire Enclosure** outage, Data remains Online and available to Hosts



HA Enclosure

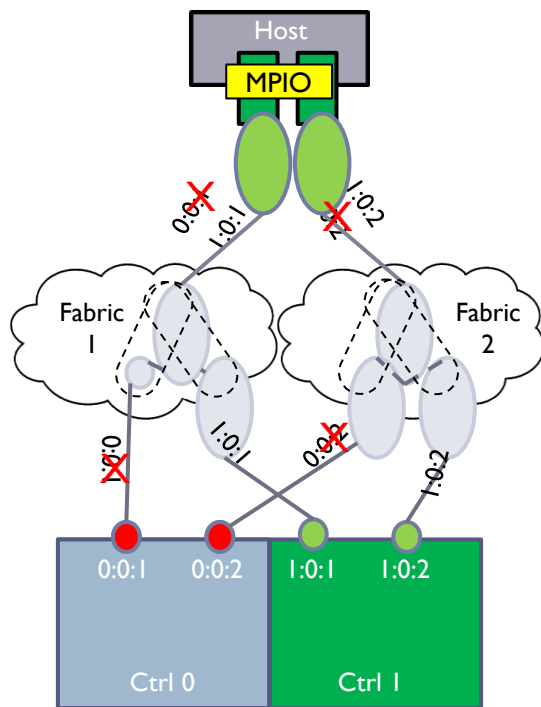
Minimum Enclosure for a given RAID size

Enclosures	RAID 1	RAID 5	RAID 6
2	OK	NA	NA
3	OK	2+1	4+2
4	OK	3+1	6+2
5	OK	4+1	8+2
8	OK	7+1	14+2



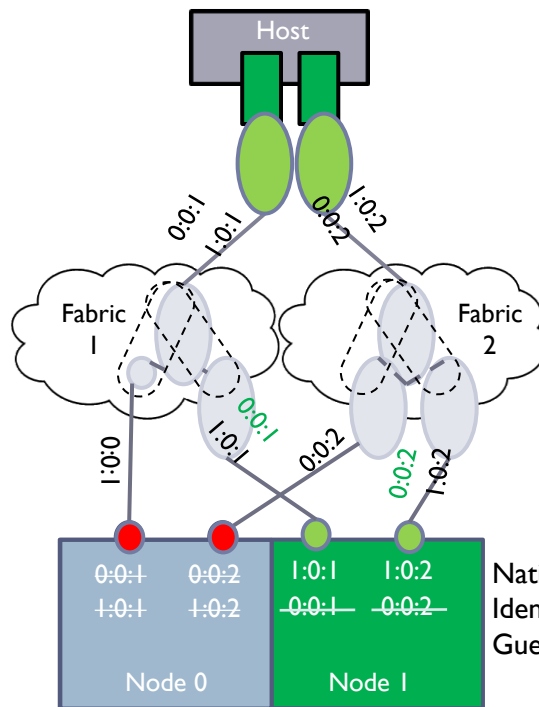
HP 3PAR Persistent Ports

Traditional Array



MPIO path failover required

HP 3PAR with Persistent Ports



All paths stay online
Node 1 takes over

- All paths stay online in case of node maintenance or failure
- No user intervention required
- NPIV based port ID swap
- Server will not “see” the swap of the 3PAR port WWN thus no MPIO path failover required



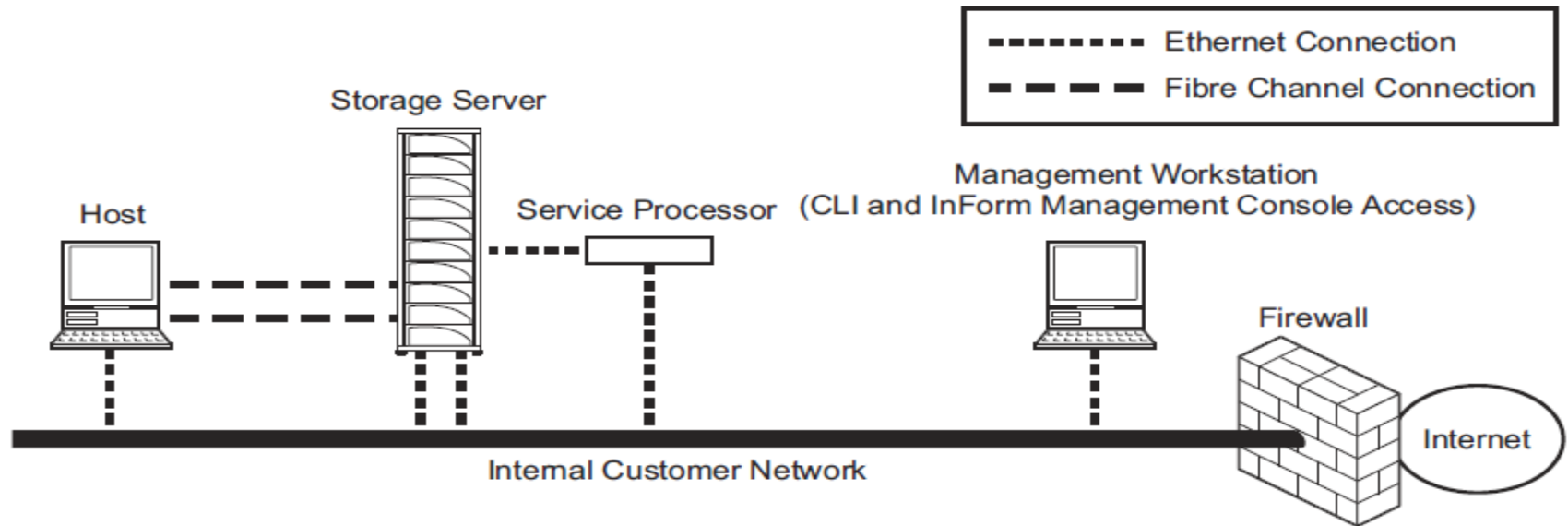
3PAR VMware Integration

HPE 3PAR is the perfect fit for virtualized environments

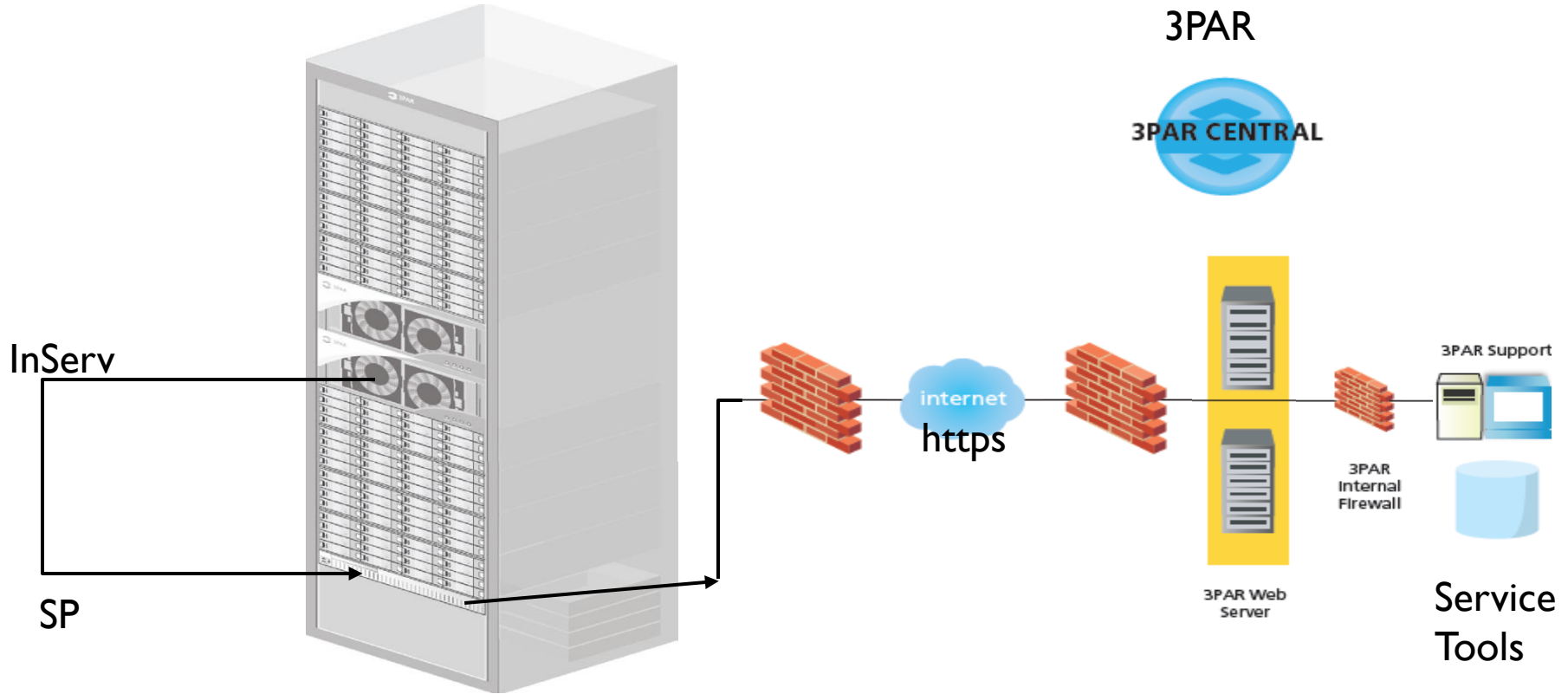
- Efficient integration of HP 3PAR Thin Technologies
- Simplified storage administration with vCenter Server and vRealize Operations Manager (vROPS ex vCOPS) integration; VAAI and VASA support ; vSphere 6.x vVol support; vCenter Site Recover Manager (SRM) integration
- High availability and disaster tolerance thanks to vSphere Metro Storage Cluster certification
- Allows greater virtual machine (VM) density thanks to inherent wide-striping and mixed workload support
- Easy recovery and replication using HP 3PAR Recovery Manager Software for VMware vSphere



3PAR Service Architecture Diagram

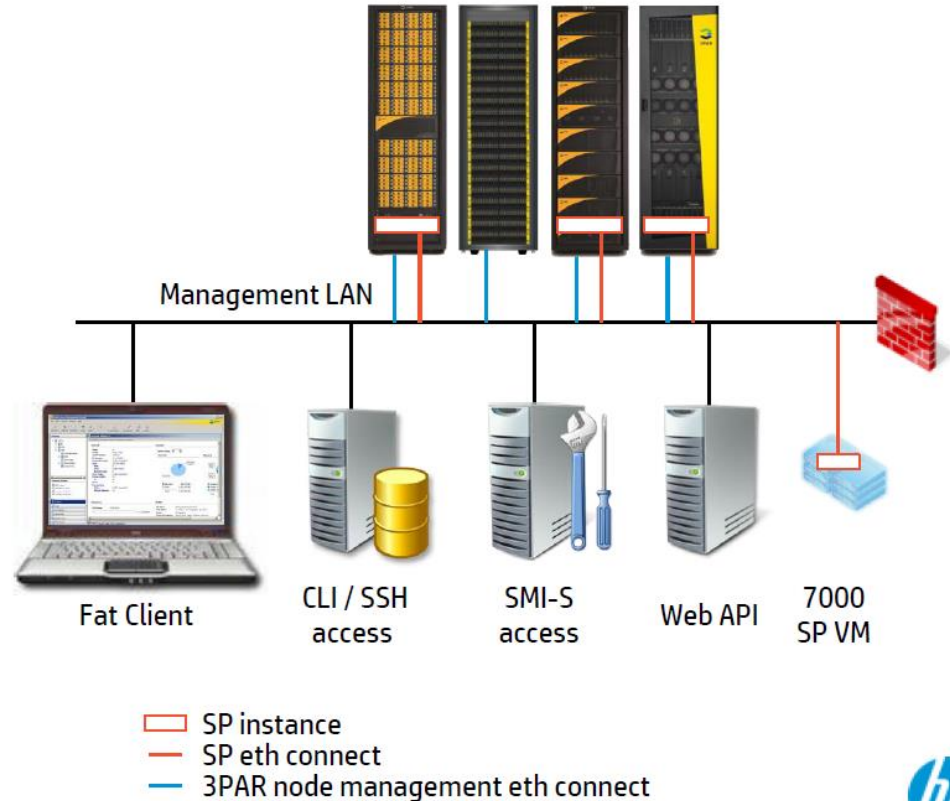


3PAR Service Architecture Diagram



HP 3PAR – Management Options

- **3PAR Management Client (GUI)**
 - Fat client GUI - Windows, RedHat Linux
 - Storage Management GUI
- **CLI**
 - 3PAR CLI or ssh
 - Storage Management Interface
 - Storage Server - very rich, complete command set
- **SMI-S**
 - Management from third party management tools
- **Web API**
 - RESTful Interface
- **Service Processor (SP)**
 - Health checks by collecting configuration and performance data
 - Reporting to HP 3PAR Central
 - Anomalies reported back to customer via OSSA
 - Array management



HP 3PAR Virtual Service Processor

Secure Remote Support

Virtual Service Processor

- Cost-efficient, secure gateway for remote connectivity
- Effortless, one-click configuration
- Supported on VMware vSphere
- Enables
 - Remote, online SW upgrade
 - Proactive fault detection with remote call home diagnostics
 - Remote serviceability
 - Alert notifications
- Optional HW Service Processor available

The screenshot shows the 'Service Processor Setup Wizard' window, specifically the 'Configure Remote Support' step. On the left, a 'Steps' list shows: 1. Welcome, 2. SP Networking, 3. Remote Support (highlighted), 4. Time and Region, 5. Change Password, 6. Summary, 7. Apply Settings, and 8. Finish. The main content area explains that Remote Support sends diagnostic information to HP 3PAR Central for analysis and proactive fault detection. It lists the diagnostic data sent: System health statistics, Configuration data, Performance data, and System events. A note states that all remote communications are encrypted and transferred securely to HP 3PAR Central, and no customer application data is ever transferred. It then lists the support provided: Remote online upgrades, Guided maintenance updates, and Remote troubleshooting. A note at the bottom states that the wizard will enable Remote Support upon completion and refers to the HP 3PAR StoreServ 7000 Storage Installation Guide. Below this, there are configuration options: 'Enable proxy server for remote support' (unchecked), with fields for Protocol (http), Address, and Port; 'Enable authentication' (unchecked), with fields for Login ID, Password, and Re-enter Password; and an 'Advanced' section with 'Make contents of Service Processor log files anonymous' (unchecked). At the bottom right are navigation buttons: '< Prev', 'Next >', 'Finish', and 'Cancel'.

Service Processor Setup Wizard

Steps

1. Welcome
2. SP Networking
3. Remote Support
4. Time and Region
5. Change Password
6. Summary
7. Apply Settings
8. Finish

Configure Remote Support

Remote Support sends diagnostic information that enables HP to perform remote analysis and proactive fault detection on your HP 3PAR StoreServ Storage system. The diagnostic information sent includes:

- System health statistics
- Configuration data
- Performance data
- System events

All remote communications are encrypted and transferred securely to HP 3PAR Central. **No** customer application data is ever transferred to HP 3PAR Central.

Remote Support provides you the best possible support for your HP 3PAR StoreServ Storage system, including:

- Remote online upgrades to your software, saving you time and money
- Guided maintenance updates, which ensures timely and reliable service
- Remote troubleshooting, which minimizes the impact of any issues to your business

This wizard will enable Remote Support upon completion.
For more information, see the *HP 3PAR StoreServ 7000 Storage Installation Guide*.

☐ Enable proxy server for remote support

Protocol: Address: Port:

☐ Enable authentication

Login ID:
Password:
Re-enter Password:

Advanced

☐ Make contents of Service Processor log files anonymous ⓘ

< Prev Next > Finish Cancel

**Install Complete**

- ✔ Introduction
- ✔ Choose Install Folder
- ✔ Pre-Installation Summary
- ✔ Installing...
- ✔ Logout Recommended
- ➔ Install Complete

Congratulations! HP 3PAR Management Console 4.3 has been successfully installed to:

C:\Program Files (x86)\3PAR\inform_mgmt_console

Press "Done" to quit the installer.

InstallAnywhere

Cancel

Previous

Done

HP 3PAR Management Console

File View Actions Window Help

Connect... Refresh Maximize Home Back Forward Export Data...

Systems

Storage Systems

3parSentry

Controller Nodes

Ports

Drive Cages

Physical Disks

FC

NL

SSD

Common Actions

Edit System...
Configure CNA Port...
Configure FC Port...
Configure iSCSI Port...
Configure RCIP Port...

Systems

Hosts

Provisioning

Connected to 172.23.1.101

Systems : Storage Systems

Introduction Summary Alerts

System Manager

HELP TOPICS

System Summary »

Capacity Information »

Port Configuration »

Software Version Information »

Health

Performance

Capacity Utilization

The **System Manager** within the HP 3PAR Management Console provides you with a high level-view of the health, performance, and capacity utilization of all your HP 3PAR Storage Systems from a single screen. From the System Manager screen, you can view statistics and drill down to access the other managers within the HP 3PAR Management Console, such as the Host Manager. You can also perform basic configuration operations from the System Manager, such as configuring ports.

The tightly-clustered HP 3PAR Architecture brings together both hardware and software components to deliver simplicity and efficiency by eliminating the price premiums and scaling complexities associated with monolithic and modular storage arrays. All HP 3PAR storage system models leverage this

New Alerts Recent Tasks My Connections

System	ID	Type	Name	Status	User	Progress	Start Time	Finish Time	Duration	Priority
--------	----	------	------	--------	------	----------	------------	-------------	----------	----------

HP 3PAR Management Console

HP 3PAR Management Console

FileViewActionsWindowHelp

Connect...RefreshMaximizeHomeBackForwardExport Data...Edit...

Systems

Storage Systems

3parSenthil

Controller Nodes

Ports

Disk

Free

Remote Copy

Drive Cages

Physical Disks

FC

NL

SSD

Common Actions

Edit System...

Configure CNA Port...

Configure FC Port...

Configure iSCSI Port...

Configure RCIP Port...

Systems

Hosts

Provisioning

Systems : Storage Systems : 3parSenthil : Drive Cages

SummaryPower SuppliesSFPsAlerts

Default8 objectsFilterClear

Any column contains:

Clear

Cage	ID	State	Model	AC State	DC State	Fan 0 State	Fan 0 Speed	Fan 1 State	Fan 1 Speed
cage0	0	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage1	0	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage2	0	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage3	0	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage0	1	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage1	1	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage2	1	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal
cage3	1	Normal	--	Normal	Normal	Normal	Normal	Normal	Normal

New AlertsRecent TasksMy Connections

System	ID	Type	Name	Status	User	Progress	Start Time	Finish Time	Duration	Priority
--------	----	------	------	--------	------	----------	------------	-------------	----------	----------

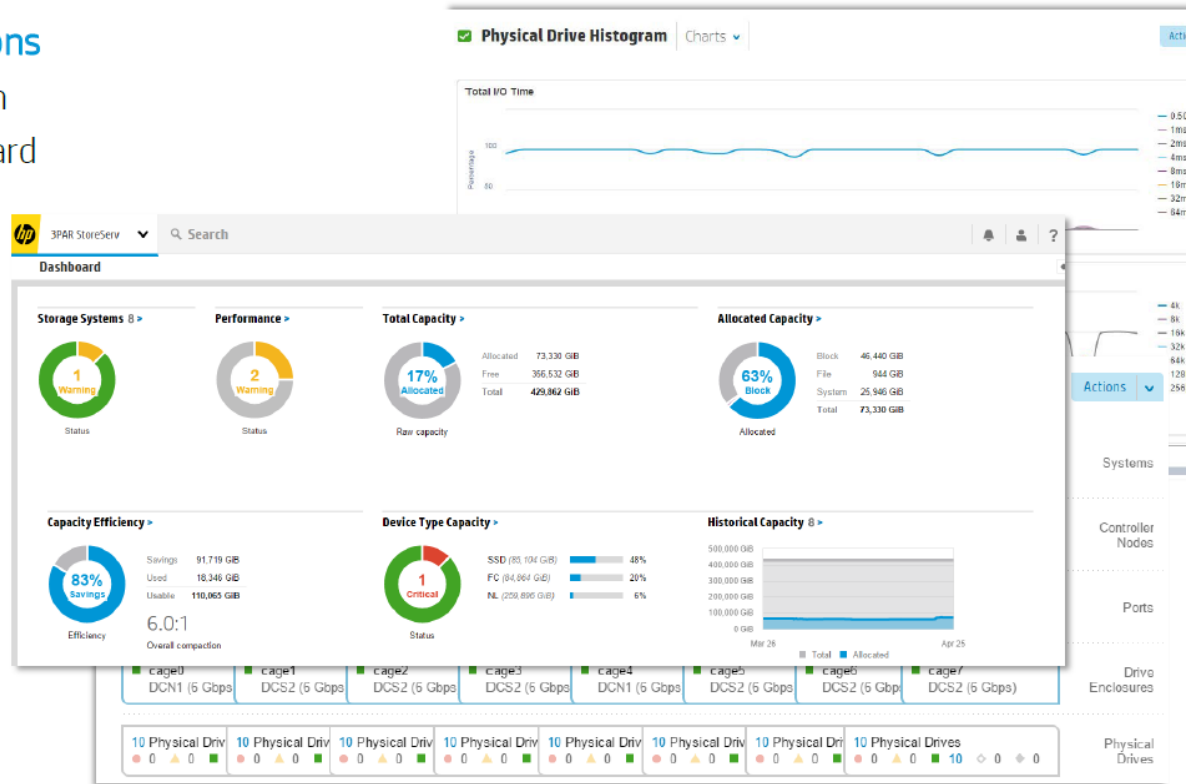
Connected to 172.23.1.101

```
Using username "3paradm".
Using keyboard-interactive authentication.
Password:
3parSenthil cli%
3parSenthil cli%
3parSenthil cli% showcage
Id Name  LoopA Pos.A LoopB Pos.B Drives Temp RevA RevB Model Side
0 cage0 1:2:1 0 0:2:1 0 12 20-20 3211 3211 DCS2 n/a
1 cage1 1:0:1 0 0:0:1 0 12 20-20 3211 3211 DCN1 n/a
2 cage2 1:1:1 0 0:1:1 0 12 20-20 3211 3211 DCS2 n/a
3 cage3 1:4:1 0 0:4:1 0 12 20-20 3211 3211 DCS1 n/a
3parSenthil cli% █
```

HP 3PAR StoreServ Management Console

Definitive Advantages.. Customer Reactions

- Up to 16 arrays managed from one screen
- Instant display of array(s) status -dashboard
- Compact Screen – same as OneView
- Single Login
- Login Intelligent
- Search
- Mapping
- Capacity Planning
- Reporting
- Easy Menu to understand



Storage Systems

Dashboard

Storage Systems 8



Performance



Systems 8 All systems ▾ All statuses ▾ Reset

Name	Model	Serial Number	HP 3PAR OS Version	Up Since	Summary ▾
mktg-eos64	HP_3PAR 7400	1645549	3.2.1.200 (MU2)	22-Dec-2014 10:53:07 o'clock P ST	
mktg-eos62	HP_3PAR 7200	1645513	3.2.1.200 (MU2)	22-Dec-2014 11:17:57 o'clock P ST	
mktg-eos67	HP_3PAR 7450	1645347	3.2.1.200 (MU2)	22-Apr-2015 17:30:19 o'clock P DT	
ebc_7450c10	HP_3PAR 7450c	1648622	3.2.1.222 (P108)	23-Feb-2015 14:08:21 o'clock P ST	
mktg-eos65	HP_3PAR 7400	9904067	3.2.2.128 (DEVEL)	09-Apr-2015 16:05:06 o'clock P DT	
Discover7450	HP_3PAR 7450	1645333	3.2.1.236 (P12)	21-Apr-2015 11:35:58 o'clock P DT	
mktg-ins3	InServ V400	1405338	3.2.1.200 (MU2)	02-Jun-2014 08:30:01 o'clock P DT	
B4_7450	HP_3PAR 7450	1699572	3.2.1.212-20150202.101907 (MU3)	15-Apr-2015 09:38:15 o'clock P DT	

mktg-eos62 View
Overview
Configuration
Capacity
Compaction
System Reporter
Settings
Software
Licenses
Activity
Map

Diagram showing the physical layout of mktg-eos62. It includes controller nodes (1645513-1), ports (SAS (Disk) 6 Gbps), drive enclosures (cage0, cage3), and physical drives (10 Physical Drives each).

Capacity Efficiency >



Savings 91,719 GiB
Used 18,346 GiB
Usable 110,065 GiB

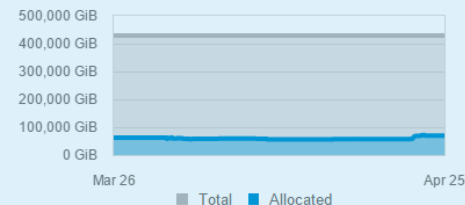
6.0:1
Overall compaction

Device Type Capacity >



SSD (85,104 GiB) 48%
FC (84,864 GiB) 20%
NL (259,896 GiB) 6%

Historical Capacity >



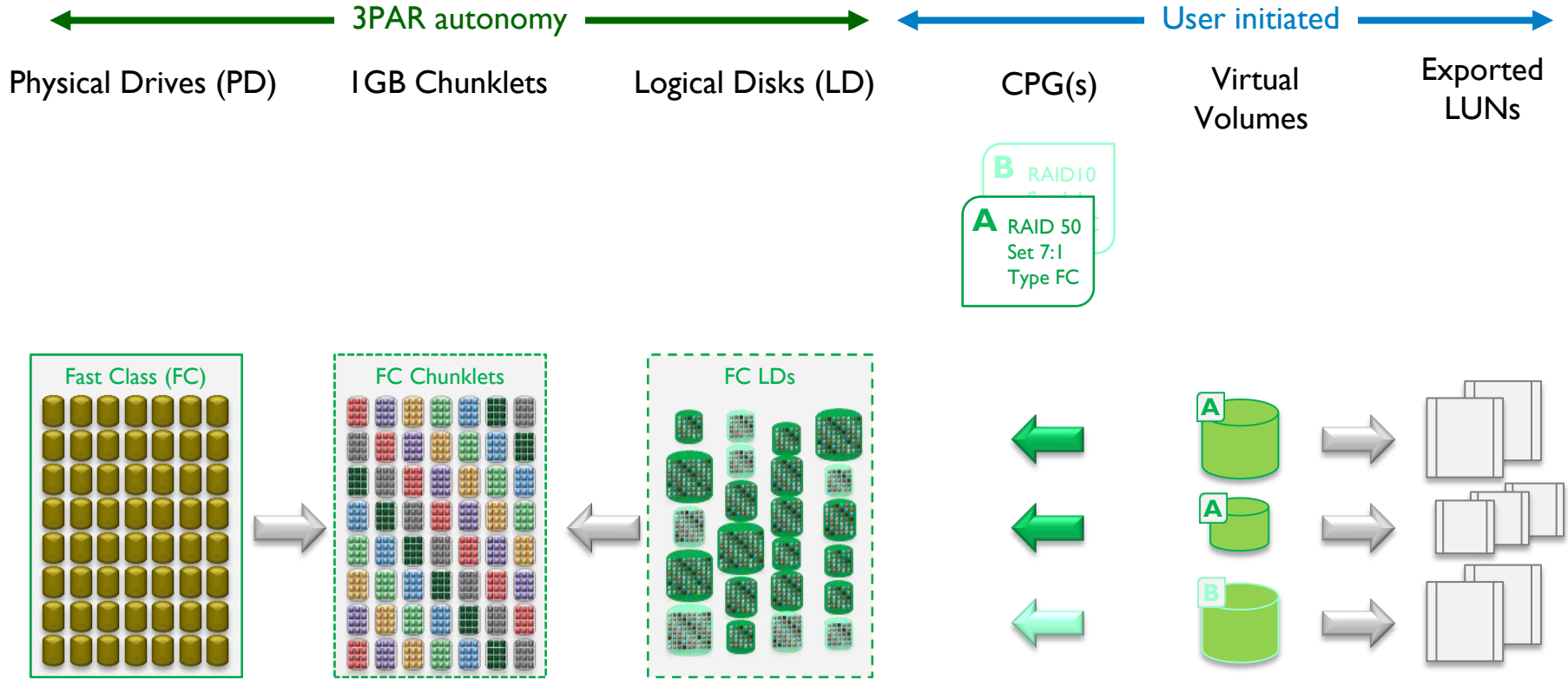
A dark blue vertical bar is positioned on the left side of the slide, spanning the height of the main content area.

HP 3PAR OS™ Virtualization Concepts

A light blue vertical bar is positioned on the left side of the slide, spanning the height of the footer area.

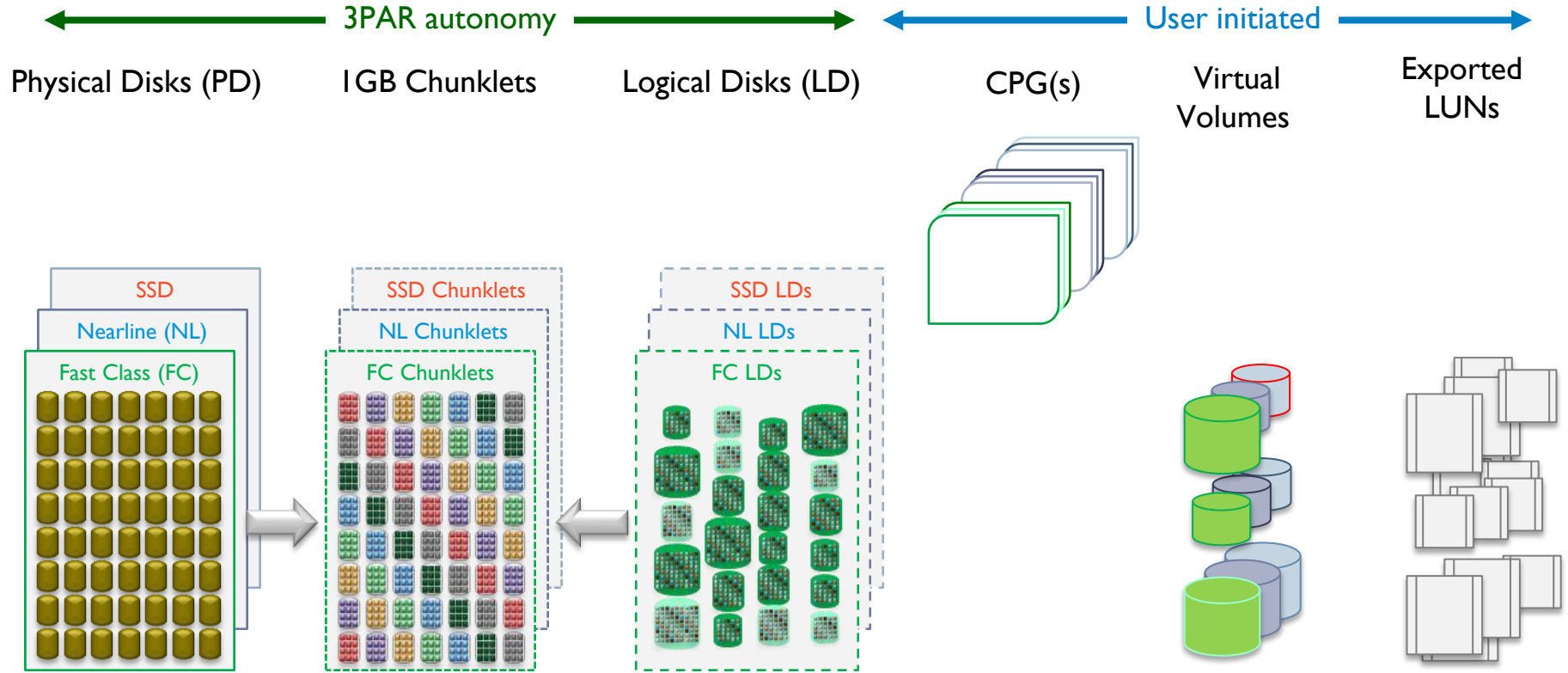
HP 3PAR Virtualization – the Logical View

With one drive type



HP 3PAR Virtualization – the Logical View

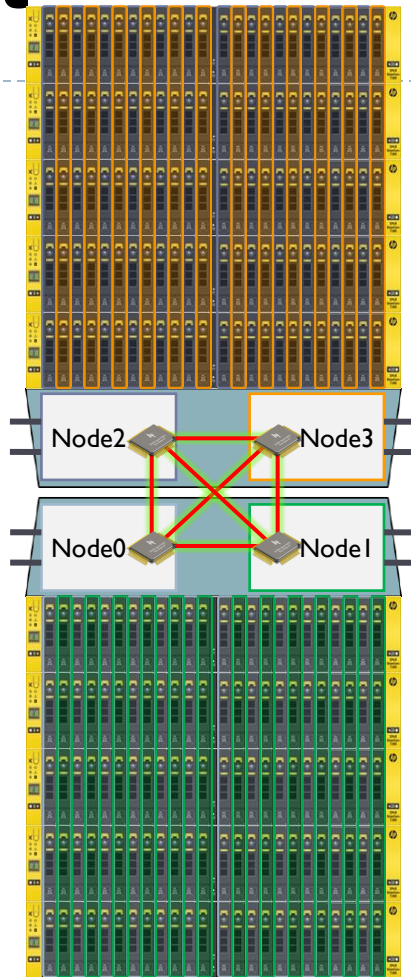
With one drive type



HP 3PAR Virtualization Concept

Example 2: 4-Node 7400 with 10 drive enclosures

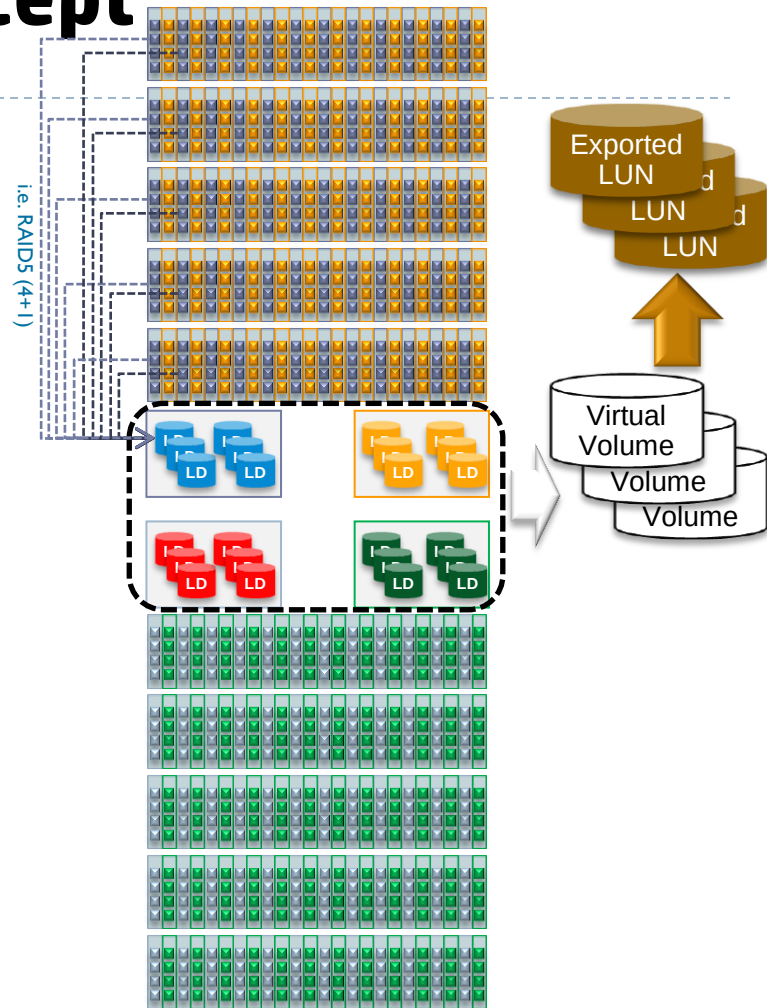
- Nodes are added in pairs for cache redundancy
Note: The nodes are installed in the back of the first drive enclosures
- 3PAR StoreServ arrays with 4 or more nodes supports “Cache Persistence”
- In this example we have a 4-node configuration with 10 drive enclosures in total
- Physical drives (PD) are owned by one node



HP 3PAR Virtualization Concept

Example: 4-Node 7400 with 10 drive enclosures

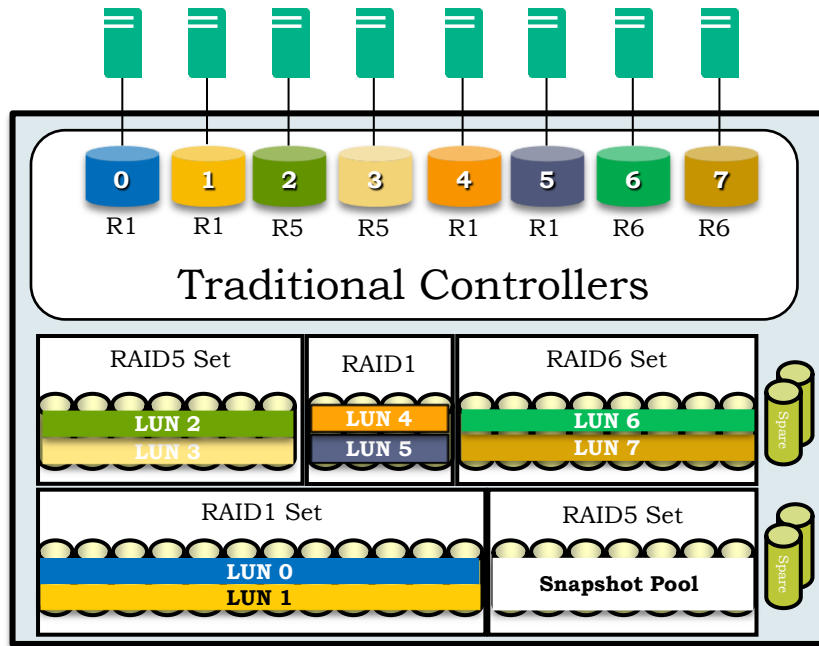
- | Process step | Phase state |
|---|--|
| <ul style="list-style-type: none">Physical drives (PD) are automatically formatted in 1GB chunklets | Disk initialization |
| <ul style="list-style-type: none">Chunklets are bound together to form Logical Disks (LD) in the format defined in CPGs | Defines RAID level, step size, set size and redundancy |
| <ul style="list-style-type: none">Virtual Volume are built on LDs across all nodes | Autonomical wide-striping across LDs |
| <ul style="list-style-type: none">Virtual Volumes can now be exported as LUNs to servers | Present LUNs ready to go |



3PAR Virtualization Advantages

Traditional Array

- Each RAID level requires dedicated drives
- Dedicated spare disk required
- Limited single LUN performance
- Dedicated pool drives



3PAR StoreServ

- All RAID levels can reside on same drives
- Distributed sparing, no dedicated spare drives
- Built-in wide-striping based on Chunklets
- No pools and reservation required

